



Flash QSPI Controller IP

Part Number: IP6514E

NDA # _____

Engineering Datasheet

Version 1.9

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1. General Information

Cadence Flash QSPI Controller IP and Cadence Flash QSPI Controller and PHY IP

This engineering datasheet describes two IP designs – a controller-only solution and a controller combined with a PHY module, both of them jointly referred to in this document as the Cadence Flash QSPI Controller IP. The Flash QSPI Controller IP can be used to provide access to Serial Flash devices. Standard Serial Peripheral Interface (SPI) is supported along with high performance Dual and Quad SPI variants.

The Cadence Flash QSPI Controller IP, subsequently referred to in this document also as the “Controller”, can be delivered as a controller-only solution (IP6514E) or as a controller fully integrated with a PHY module (IP6515E).

The Cadence Flash QSPI Controller IP connects to a system-on-chip (SoC) environment through its ARM® AMBA® AHB® bus and APB bus interfaces. The AHB interface is used to transfer data, either in a memory mapped direct fashion (for example a processor wishing to execute code directly from external FLASH memory), or in an indirect fashion where the controller is setup via configuration registers to silently perform some requested operation, signalling its completion via interrupts or status registers.

For indirect operations, data is transferred between system memory and external FLASH memory via an internal SRAM which is loaded for writes and unloaded for reads by an AHB master within the SoC environment at low latency AHB system speeds. Interrupts or status registers are used to identify the specific times at which this SRAM should be accessed using user programmable configuration registers. The size of the SRAM is configurable.

An optional DMA peripheral bus is also available to optimize the data transfers between an external master and the Cadence Flash QSPI Controller IP during indirect transfers.

1.1 Key Features

The Cadence® Flash QSPI Controller IP supports the following key features:

- Memory mapped ‘direct’ mode of operation for performing FLASH data transfers and executing code from FLASH memory
- Software triggered ‘indirect’ mode of operation for performing low latency and non-processor intensive FLASH data transfers
- Optional DMA Peripheral interface to communicate indirect mode status with external DMA
- Local SRAM of configurable size to reduce AHB overhead and buffer FLASH data during indirect transfers
- Set of software APB accessible FLASH control registers to perform any FLASH command, including data transfers up to 8-bytes at a time
- Supports any device clock frequency, including current market device frequencies of 133MHz SDR or 80 MHz DDR
- Support for XIP (Execute in Place), sometimes referred to as continuous mode
- Support for DDR Mode and DTR Protocol
- Support for single, dual or quad I/O instructions
- Programmable device sizes
- Programmable write protected regions to block system writes from taking effect

- Programmable delays between transactions
- Legacy mode allowing software direct access to low level transmit and receive FIFOs, bypassing the higher layer processes
- Option to use an independent reference clock to decouple AHB clock from SPI clock – allows slow system clocks
- Serial clock with programmable polarity
- Programmable baud rate generator to generate device clocks
- Features included to improve high speed read data capture mechanism
- Option to use adapted clocks to further improve read data capturing
- Programmable interrupt generation
- Up to four external device selects
- Programmable AHB decoder, enables continuous addressing mode for each of connected devices and auto-detection of boundaries between devices
- Support BOOT mode

Optionally, you can get full integration with a PHY module dedicated to more flexible and power efficient transfers (IP6515, the Cadence Flash QSPI Controller IP integrated with a PHY IP).

1.2 Benefits

The Cadence Flash QSPI Controller IP has the following benefits:

- Low-risk solution with silicon-proven design
- Ease-of-use — customizable with easy integration
- Efficiency — optimized data transfer ensured

2. Top-Level Architecture

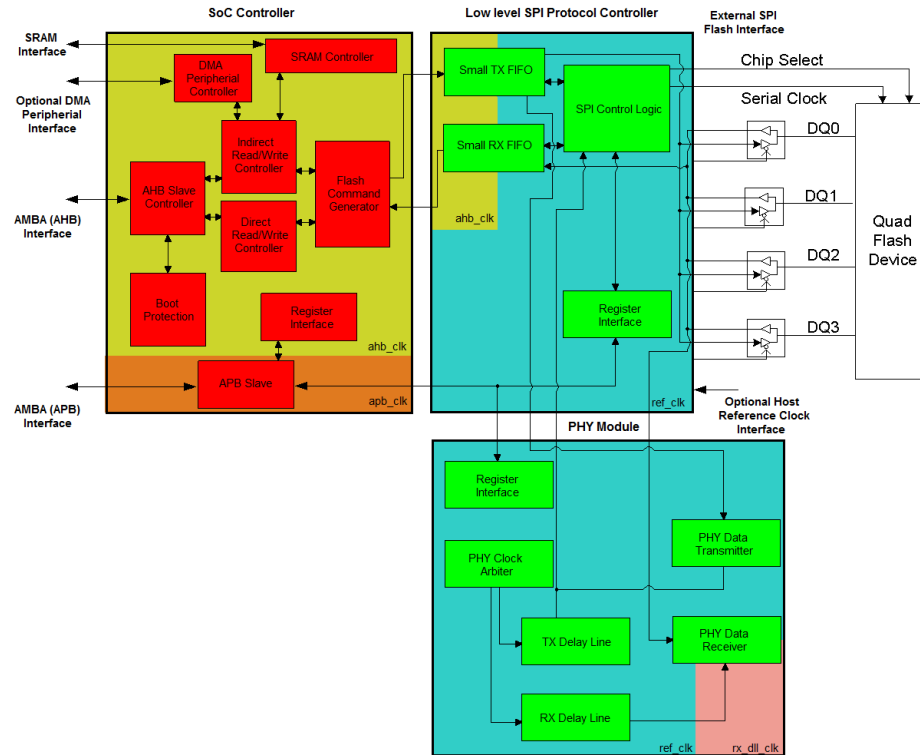


Figure 1 Controller Block Diagram

Figure 1 shows the architecture of the Flash QSPI Controller IP with the optional PHY Module.

3. Signal Interfaces

The Flash QSPI Controller IP includes the following signal interfaces:

- AMBA Advanced Peripheral Bus (APB) slave interface for accessing the programmable registers
- AMBA Advanced High Speed Bus (AHB) slave interface for data transfer
- An optional DMA peripheral interface

The AMBA interfaces fully conform to the ARM AMBA 2 Specification.

Table 1 AMBA (APB) Interface

Signal Name	I/O	Description
n_preset	I	Active low AMBA reset (nPRESET). This signal must be asserted low asynchronously, and deasserted high synchronously with pclk. Resets all APB registers.
pclk	I	Peripheral bus clock (PCLK).
psel	I	Peripheral select (PSEL). Active high select signal to indicate that a valid access is being made to one of the programmable registers.
penable	I	Peripheral enable (PENABLE). This indicates the second clock cycle of an access and indicates that the write data may be strobed into a register on the next rising edge of pclk, or that the read data is expected to be valid at the next rising edge of pclk.
pwrite	I	Peripheral write strobe (PWRITE). This indicates that a write access is taking place (if psel is active).
paddr[7:0]	I	Address bus from selected master (PADDR). Indicates which register is being accessed. This address is word-aligned and therefore bits [1:0] should always be '00'.
pwwdata[31:0]	I	Write data. Data to be written into the addressed register.
prdata[31:0]	O	Read data. Data read from the addressed register. For APB Rev 2.0, it is driven to logic 0 when not addressed.
pslverr	O	Indicates an APB access to an invalid address. Should be sampled when penable is high. This signal is optional and not a standard AMBA signal.
pready	O	Indicates the current APB access is to be delayed with wait states. If pready is tied high – i.e. no wait states.
interrupt	O	Level sensitive active high interrupt.

Table 2 AMBA (AHB) Interface

Signal Name	I/O	Description
n_hreset	I	Active low asynchronous reset. This signal must be asserted low asynchronously and deasserted high synchronously with hclk.
hclk	I	AHB bus clock.
hsel	I	AHB slave select.
hready_in	I	Combined hready from all slaves in AHB system.
htrans[1:0]	I	AHB transfer type as follows: 00 - Idle 01 - Busy 10 - Non-sequential 11 - Sequential.
haddr[31:0]	I	AHB address.
hwrite	I	High for AHB write, low for AHB read.
hsize[2:0]	I	Transfer size, configured according to AMBA AHB data bus width as follows: 010 - 32 bit transfers 011 - 64 bit transfers 100 - 128 bit transfers.
hburst[2:0]	I	Burst type, 000 for single transfer, 001 for incrementing burst and 011 for four beat incrementing burst, etc.
hwdata[31:0]	I	Write data from external AHB Master, 32-bit only.
hrdata[31:0]	O	Read data to external AHB Master, 32-bit only. Unused bits will be tied to logic 0.
hresp	O	Slave response, 0 for OK, any other response is not OK and triggers an interrupt. AMBA split and retry operations are not supported.
hready_out	O	hready generated by the Flash Controller slave. When low, the flash controller is inserting wait states.

Table 3 Optional Host Reference Clock Interface

Signal Name	I/O	Description
n_ref_rst	I	Active low ref_clk reset. This signal must be asserted low asynchronously, and deasserted high synchronously with ref_clk. Resets all registers in the reference clock domain.
ref_clk	I	Reference clock (ref_clk).

The SRAM ports (used for indirect modes of data transfer only) are passed to the top level

Table 4 SRAM Interface

Signal Name	I/O	Description
sram_cs	O	Active high chip select.
sram_we	O	Active high write enable.
sram_re	O	Active high read enable.
sram_add[N:0]	O	SRAM address bus. Address width configurable.
sram_do[31:0]	O	SRAM write data bus.
sram_di[31:0]	I	SRAM read data bus.

Table 5 External SPI Flash interface

Signal Name	I/O	Description
mi0	I	Quad-SPI master mode input, bit 0 The master input may be up to 4-bits wide in master mode, with 1-bit for standard serial SPI operation, 2-bits for Dual-SPI, and 4-bits for Quad-SPI Internally, the Quad-SPI IP must switch between mi[1] for 1-bit wide input and mi[0] for the LSB of dual-mode or quad-mode input
mo0	O	MOSI signal, master mode output; Quad-SPI master mode output, bit 0
n_mo_en[0]	O	Master mode output enable for bit 0. Active low. This pin retains its original function for serial SPI mode, but additionally will be driven dynamically for Dual- and Quad-SPI modes, based on the enabled mode and the current stage of the transfer.
mi1	I	MISO pin, master mode input; Quad-SPI master mode input, bit 1 This may be up to 4-bits wide in master mode, with 1-bit for standard serial SPI operation, 2-bits for Dual-SPI, and 4-bits for Quad-SPI Internally, the Quad-SPI IP must switch between mi[1] for 1-bit wide input and m[0] for the LSB of dual-mode or quad-mode input
mo1	O	Quad-SPI master mode output, bit 1
n_mo_en[1]	O	Slave mode output enable; Master mode output enable for bit 1 Active low. This signal supersedes and replaces the n_so_en signal from the Original SPI core.
mi2	I	Quad-SPI master mode input, bit 2 This may be up to 4-bits wide in master mode, with 1-bit for standard serial SPI operation, 2-bits for Dual-SPI, and 4-bits for Quad-SPI
mo2_wpn	O	Quad-SPI master mode output, bit 2, for quad-mode writes to external SPI slave and used for write protect output to flash devices, active low, when not in quad mode.

Signal Name	I/O	Description
n_mo_en[2]	O	Master mode output enable for bit 2. Active low. Only utilized if quad-mode commands are supported.
mi3	I	Quad-SPI master mode input, bit 3 This may be up to 4-bits wide in master mode, with 1-bit for standard serial SPI operation, 2-bits for Dual-SPI, and 4-bits for Quad-SPI
mo3_hold	O	Quad-SPI master mode output, bit 3, for quad-mode writes to external SPI slave, and used for hold output to flash devices, active low, when not in quad mode. NOTE: The use of the HOLD pin is not supported by the controller.
n_mo_en[3]	O	Master mode output enable for bit 3. Active low. Only utilized if quad-mode commands are supported.
n_ss_out[3:0]	O	Quad-SPI peripheral select outputs The IP supplies up to 4 CS
sclk_out	O	Quad-SPI master clock output

Table 6 Optional DMA Peripheral Interface

Signal Name	I/O	Description
indwr_drtype	O	Driven by controller to indicate the request type the indirect write logic wants the dma to perform (single or burst)
indwr_drlast	O	Indirect write dma request last
indwr_drvalid	O	Indirect write controller request to dma to perform a transfer
indwr_drready	I	Indication that dma has accepted the indirect write dma request
indwr_datype	I	Driven by dma to indicate to the controller the type of transfer that is being acknowledged. The QSPI only uses this signal to identify an incoming DMA flush request.
indwr_davalid	I	Driven by dma to indicate that the dma request is complete
indwr_daready	O	Driven by controller to indicate that it saw the indwr_davalid
indrd_drtype	O	Driven by controller to indicate the request type the indirect read logic wants the dma to perform (single or burst)
indrd_drlast	O	Indirect read dma request last
indrd_drvalid	O	Indirect read controller request to dma to perform a transfer
indrd_drready	I	Indication that dma has accepted the indirect read dma request
indrd_datype	I	Driven by dma to indicate to the controller the type of transfer that is being acknowledged. The QSPI only uses this signal to identify an incoming DMA flush request.
indrd_davalid	I	Driven by dma to indicate that the indirect read dma request is complete

Signal Name	I/O	Description
indrd_daready	O	Driven by controller to indicate that it saw the indrd_davalid

Table 7 Adapted Loopback Device Clock

Signal Name	I/O	Description
sclk_dlyd_i	I	This is an optional clock, used by the read delay capture circuit to sample the read data as it returns from the memory device. It is a delayed version of the 'sclk_out' output from the QSPI IP. The delay itself is implemented externally. Refer to the UserGuide for further details.

4. Timing Requirements

4.1 APB Timings

Table 8 APB Timings

Parameter	Description	Min	Max	Unit
Tclk	Clock period	10	DC	ns
Tclkh	Clock high	N.A.	N.A.	
Tclkl	Clock low	N.A.	N.A.	
Tihnres	n_preset hold after pclk	N.A.	N.A.	
Tisnres	n_preset setup before pclk	N.A.	N.A.	
Tihpen	penable hold after pclk	0.5	-	ns
Tispen	penable setup before pclk	4.0	-	ns
Tihpel	psel hold after pclk	0.5	-	ns
Tispel	psel setup before pclk	4.0	-	ns
Tihpa	paddr hold after pclk	0.5	-	ns
Tispa	paddr setup before pclk	4.0	-	ns
Tihpw	pwrite hold after pclk	0.5	-	ns
Tispw	pwrite setup before pclk	4.0	-	ns
Tihpwd	pdata hold after pclk	0.5	-	ns
Tispwd	pdata setup before pclk	4.0	-	ns
Tohprd	prdata hold after pclk	0.5	-	ns
Tovprd	prdata valid after pclk	-	4.0	ns

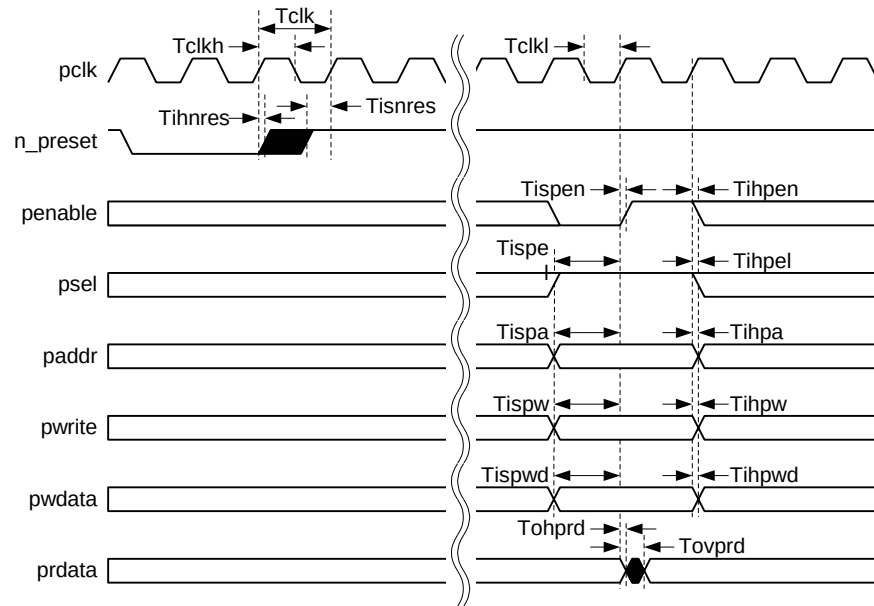


Figure 2 APB Timings

4.2 AHB Timings

Maximum achieved AHB frequency is equal to 600 MHz so this requirement was inserted into the table. There are some limitations which could hamper correct work on such high frequency depending on individual requirements of the system.

Table 9 AHB Timings

Parameter	Description	Min	Max	Unit
Tclk	Clock period, T0	1.6	DC	ns
Tclkh	Clock high	40%	—	T0
Tclkl	Clock low	40%	—	T0
Tihnres	n_sys_reset hold after hclk	N/A		T0
Tisnres	n_sys_reset setup before hclk	N/A		T0
Tissel	hsel setup before hclk	60%	—	T0
Tihsel	hsel hold after hclk	0.1	—	ns
Tohrdy	hready_out hold after hclk	0.1	—	ns
Tovrdy	hready_out valid after hclk	—	50%	T0
Tihrdy	hready_in hold after hclk	0.1	—	ns
Tisrdy	hready_in setup before hclk	60%	—	T0
Tihtr	htrans hold after hclk	0.1	—	ns
Tistr	htrans setup before hclk	60%	—	T0
Tiha	haddr hold after hclk	0.1	—	ns
Tisa	haddr setup before hclk	60%	—	T0
Tihctl	hwrite, hsize hold after hclk	0.1	—	ns
Tisctl	hwrite, hsize setup before hclk	60%	—	T0
Tihwd	hwdata hold after hclk	See Note		
Tiswd	hwdata setup before hclk	60%	—	T0
Tohrsp	hresp hold after hclk	0.1	—	ns
Tovrsp	hresp valid after hclk	—	50%	T0
Tohrd	hrdata hold after hclk	0.1	—	ns
Tovrd	hrdata valid after hclk	—	50%	T0

4.3 QSPI Timing

Most outputs are registered on the ref_clk. There is no combinatorial logic on these outputs. Master Output Interface which is directly forwarded to external Flash Device has multiplexer on its output for the optional delay inserting feature which is used to adjust hold timing. Given that this interface toggles based on programmable divider synchronous for ref_clk – multi cycle path is ensured, therefore additional combinational delay is safe in this case. For the inputs, these are all data inputs, driven on one edge of sclk_in (which itself is synchronous to ref_clk), and sampled by the QSPI flash controller on the ref_clk.

Table 10 QSPI Timings

Parameter	Description	Min	Max	Unit
INPUTS				
Tclk	ref_clk period, T0	1.25	DC	ns
Tihmi0	mi0 hold after ref_clk	5%	-	%Tclk
Tismi0	mi0 setup before ref_clk	-	35%	%Tclk
Tihmi1	mi1 hold after ref_clk	5%	-	%Tclk
Tismi1	mi1 setup before ref_clk	-	35%	%Tclk
Tihmi2	mi2 hold after ref_clk	5%	-	%Tclk
Tismi2	mi2 setup before ref_clk	-	35%	%Tclk
Tihmi3	mi3 hold after ref_clk	5%	-	%Tclk
Tismi3	mi3 setup before ref_clk	-	35%	%Tclk
OUTPUTS				
Tohmo0	mo0 hold after ref_clk	5%	-	%Tclk
Tovmo0	mo0 valid after ref_clk	-	50%	%Tclk
Tohmo1	mo1 hold after ref_clk	5%	-	%Tclk
Tovmo1	mo1 valid after ref_clk	-	50%	%Tclk
Tohmo2	mo2_wpn hold after ref_clk	5%	-	%Tclk
Tovmo2	mo2_wpn valid after ref_clk	-	50%	%Tclk
Tohmo3	mo3_hold hold after ref_clk	5%	-	%Tclk
Tovmo3	mo3_hold valid after ref_clk	-	50%	%Tclk
Tohmoen	n_mo_en hold after ref_clk	5%	-	%Tclk
Tovmoen	n_mo_en valid after ref_clk	-	50%	%Tclk
Tohsso	n_ss_out hold after ref_clk	5%	-	%Tclk
Tovsso	n_ss_out valid after ref_clk	-	50%	%Tclk

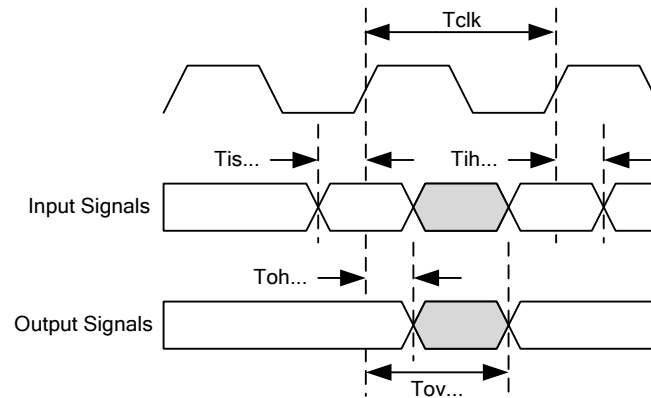


Figure 4 QSPI Timings

Note 1: Max timings are specified relative to the target clock period, Tclk. These are coded into the synthesis script provided. These timings have been achieved with a typical technology, but are for guidance only.

Note 2: Hold times specified are guidelines only. Min timings have no effect during synthesis.

4.4 Clock Domains

There are three main asynchronous clock sources for the QSPI Controller (although this can be reduced optionally to two) and one generated clock for clocking the external FLASH device.

The AHB clock is the main system clock used to transfer data over the AHB bus between an external AHB master and the QSPI controller. The APB clock is used to access the register map of the QSPI controller and perform basic configuration and interrupt handling.

The reference clock is optional via a configuration define. The reference clock (when enabled) or the AHB clock (when the reference lock is not enabled) is used to serialize the data and drive the external SPI interface. Using the reference clock allows the core to decouple the frequency of the SPI FLASH device from the SoC system clocks, thereby providing a more flexible clocking solution.

5. Programming Interface

The following registers may be programmed for the Cadence Flash QSPI Controller IP module.

Table 11 Controller Register Map

Offset	Function	R/W
0x0	QSPI Configuration Register	R/W
0x4	Device Read Instruction Register	R/W
0x8	Device Write Instruction Register	R/W
0xC	QSPI Device Delay Register	R/W
0x10	Read Data Capture Register	R/W
0x14	Device Size Register	R/W
0x18	SRAM Partition Register	R/W
0x1C	Indirect AHB Address Trigger Register	R/W
0x20	DMA Peripheral Register	R/W
0x24	Remap Address Register	R/W
0x28	Mode Bit Register	R/W
0x2C	SRAM Fill Level Register	RO
0x30	TX Threshold Register	R/W
0x34	RX Threshold Register	R/W
0x38	Write Completion Control Register	R/W
0x3C	Polling Expiration Register	R/W
0x40	Interrupt Status Register	R/W
0x44	Interrupt Mask	R/W
0x50	Lower Write Protection Register	R/W
0x54	Upper Write Protection Register	R/W
0x58	Write Protection Register	R/W
0x60	Indirect Read Transfer Register	R/W
0x64	Indirect Read Transfer Watermark Register	R/W
0x68	Indirect Read Transfer Start Address Register	R/W
0x6C	Indirect Read Transfer Number Bytes Register	R/W
0x70	Indirect Write Transfer Register	R/W

Offset	Function	R/W
0x74	Indirect Write Transfer Watermark Register	R/W
0x78	Indirect Write Transfer Start Address Register	R/W
0x7C	Indirect Write Transfer Count Register	R/W
0x80	Indirect AHB Trigger Address Range Register	R/W
0x90	Flash Command Register	R/W
0x94	Flash Command Address Registers	R/W
0xA0	Flash Command Read Data Register (Lower)	RO
0xA4	Flash Command Read Data Register (Upper)	RO
0xA8	Flash Command Write Data Register (Lower)	R/W
0xAC	Flash Command Write Data Register (Upper)	R/W
0xB0	Flash Status Register	RO
0xFC	Module ID Register	RO

6. Physical Estimates

The physical estimates listed below are applicable to the Cadence Flash QSPI Controller IP without integrated PHY IP.

Approximate two input NAND equivalent gate counts: 58000

Flip Flop count in HCLK domain (rising edge/falling edge): 2019/0

Flip Flop count in PCLK domain (rising edge/falling edge): 894/0

Flip Flop count in REF_CLK domain (rising edge/falling edge): 723/4

Flip Flop count in SCLK_DLYD_I domain (rising edge/falling edge): 0/4

7. Deliverables

Table 12 Flash QSPI Controller IP Deliverables

#	Description
1)	Clean, readable, synthesizable Verilog HDL
2)	Cadence Encounter® RTL Compiler synthesis scripts
3)	Simple Verilog testbench
4)	Documentation – integration and user guide, release notes
5)	IP-XACT xml file

A. Document Revision History

Table 13 Document Revision History

Revision	Modifications	Data	Author
1.6	Converted to the latest EDS format	14-Nov-2014	J Klimaszewska
1.7	Timing requirements were modified, Block Diagram was divided for clock domains	25-Nov-2014	M. W.
1.8	PHY Registers were added. Optional PHY Module block was added to the Block Diagram.	09-Dec-2014	M. W.
1.9	Timing requirements were modified and described more accurately	22-Dec-2014	M. W.
2.0	The document was split for two in order to created separate ones for both configurations.	06-May-2015	M. W.

